1. IPO

INPUT	PROCESSING	OUTPUT
firstNo secondno	get firstno get secondno Check smallest number from firstno and secondno as a hcf Repeat: Reduce the hcf by 1 until it divides both numbers exactly Show the hcf	Display “The Highest Common Factor is: “hcf

2. ALGORITHM

START

GET: firstno

GET: secondno

IF firstno < secondno

THEN

hcf = firstno

ELSE

hcf = secondno

ENDIF

WHILE (firstno **MOD** hcf **NOT** 0)

OR (secondno **MOD** hcf **NOT** 0)

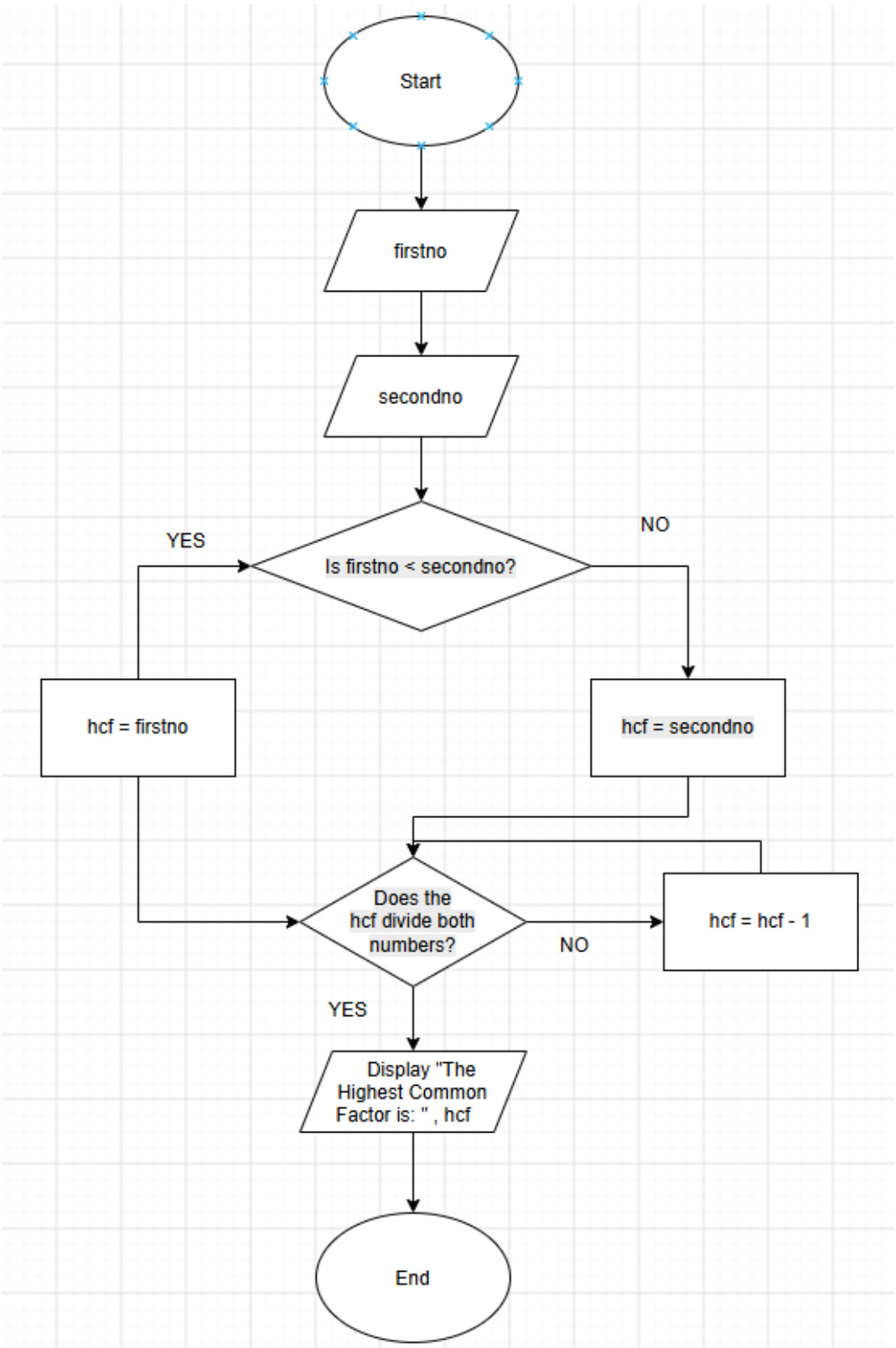
hcf = hcf – 1

ENDWHILE

DISPLAY “The Highest Common Factor is: “hcf

END

3. Flowchart


Start
firstno
secondno
Is firstno < secondno?
YES
NO
hcf = firstno
hcf = secondno
Does the hcf divide both numbers?
NO
hcf = hcf - 1
YES
Display "The Highest Common Factor is: " , hcf
End

4. Trace table

Test case	Firstno	Secondno	Expected results
1	12	37	1
2	14	20	2
3	6	18	6

